

141071 26

1021

Remove Outer most Parentheses

3

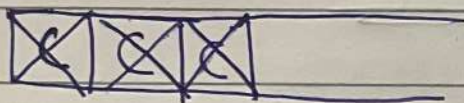
Dry Run

After completing the iteration till this

$S \rightarrow (()) (()) (()) (()) (())$ decomposition

we get

I iteration



ans $\rightarrow$ () ()

Stack

Condition:

۴۸۳۱

① Stack Empty $\rightarrow$ push into stack
 $\rightarrow$ don't add in ans

$\mathcal{E}(3 | \mathcal{E})_3 \rightarrow \text{pop to the top}$

② !Stack Empty → push int stack
→ add to ans

213

③ Stack Empty $\rightarrow$ pop from top
 $\rightarrow$ don't add in any

II

Step " (()) | (()) (()) "

completed

I iteration

~~stack~~

ans = " () () () "

stack

III

Step " (()) (()) | (() (())) "

completed

stack

ans = " () () () () () () "

ans = " () () () () () () "

Code

```
String helper (String S) {
```

```
    Stack <Character> st = new Stack<>();
```

```
    StringBuilder ans = new StringBuilder();
```

```
    for (int i = 0; i < S.length(); i++) {
```

```
        char ch = S.charAt(i);
```

```
        if (ch == '(') {
```

```
            if (st.isEmpty()) {
```

```
                st.push(ch);
```

```
            } else {
```

```
                st.push(ch);
```

```
                ans.append(ch);
```

```
            }
```

```
        } else {
```


* st.pop();

~~if (st.empty())~~

if (!st.empty()) {

ans.append(ch);

2 2

2

2 return ans.toString();

2